COMSATS University Islamabad Wah Campus



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Internship Report (Week No. 6)

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Submission Date : 14 September 2026 (Monday)

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Part 1 — Data Center Visit

1.1 Objective

The objective of this visit was to gain practical knowledge about a real-world Data Center environment and understand how different types of servers, networking equipment, storage systems, and database servers are installed and managed.

The visit also aimed to understand the practical role of a Database Administrator (DBA), particularly the management and operation of an Oracle Database Server in an organizational environment.

1.2 Introduction to Data Center

A Data Center is a specialized facility used to house and manage computer systems, servers, networking equipment, storage devices, and other IT infrastructure.

Data Centers provide the necessary environment for organizations to run important IT services and applications. They are designed to provide reliable power, cooling, networking, physical security, and controlled access to ensure that servers and services remain available.

During the visit, the instructor explained the purpose of the Data Center and provided an overview of the different types of equipment installed there.

1.3 Purpose of the Visit

The main purpose of the Data Center visit was to provide practical exposure to the infrastructure that supports organizational IT services.

During the visit, different types of servers were observed and their basic functions were explained. The instructor also explained the working and importance of an Oracle Database Server.

The visit helped us understand how the concepts learned during the DBA internship are implemented in a real-world environment.

1.4 Types of Servers Observed

During the Data Center visit, different types of servers were installed and their functions were discussed.

1.4.1 Database Server

A Database Server is used to store, manage, and process organizational data. It runs database management software such as Oracle Database and provides controlled access to data for authorized applications and users.

1.4.2 Application Server

An Application Server provides the environment required to run business applications and application-related services. It can communicate with the database server to store and retrieve application data.

1.4.3 Web Server

A Web Server provides web-based services and responds to requests received from users and client applications through a network.

1.4.4 File Server

A File Server is used to centrally store and manage files and documents. Authorized users can access shared files through the network.

1.4.5 Backup Server

A Backup Server is used to maintain backup copies of important organizational data. Backups are important for recovering data in case of accidental deletion, hardware failure, corruption, or other problems.

1.5 Oracle Database Server

One of the important servers observed during the Data Center visit was an **Oracle Database Server**.

An Oracle Database Server is a system on which Oracle Database software is installed and running. It provides centralized storage, processing, and management of organizational data.

The instructor explained that the Oracle Database Server handles database requests, stores data, manages database users and privileges, and provides database services to authorized applications and users.

This was directly related to the practical Oracle Database administration activities performed during the internship.

1.6 Working of Oracle Database Server

The basic working of an Oracle Database Server can be summarized as follows:

1. A user or application sends a request to the Oracle Database.
2. The request is received through the appropriate database service.
3. Oracle processes the SQL statement or database operation.
4. The database accesses the required data from storage.
5. Oracle performs the requested operation.
6. The result is returned to the user or application.

For remote database connections, the **Oracle Listener** plays an important role. It receives connection requests and directs them to the appropriate Oracle Database service.

The Database Administrator monitors the database, listener, storage, users, security, and performance to ensure that the database remains available and operates correctly.

1.7 Role of DBA in Data Center

A Database Administrator has an important role in maintaining and managing database infrastructure in a Data Center.

The major responsibilities of a DBA include:

- Installing and configuring databases
- Starting and shutting down database services
- Managing database users and privileges
- Maintaining database security
- Monitoring database performance
- Managing tablespaces and storage
- Performing backup and recovery operations
- Managing database connectivity
- Monitoring Oracle Listener and database services
- Troubleshooting database errors
- Monitoring overall database health
- Maintaining database availability

The DBA ensures that the database remains secure, available, reliable, and capable of supporting organizational applications.

1.8 Database Infrastructure and Connectivity

Database infrastructure consists of the hardware, software, networking, storage, and supporting components required to operate a database system.

In an Oracle Database environment, the basic connectivity can be represented as:

Client/Application → Network → Oracle Listener → Database Service → Oracle Database → Storage

When a client such as SQL Developer or TOAD attempts to connect to an Oracle Database, the connection request is sent through the network to the Oracle Listener.

The Listener identifies the requested database service and forwards the connection to the appropriate database instance or service.

Proper configuration of the **IP address, hostname, port, listener, service name, and TNS configuration** is important for successful database connectivity.

1.9 Learning Outcomes

The Data Center visit provided the following learning outcomes:

- Developed an understanding of a real-world Data Center environment.
- Observed different types of servers and their functions.
- Understood the purpose and importance of an Oracle Database Server.
- Learned the basic working of an Oracle Database Server.
- Understood the role of an Oracle Listener in database connectivity.
- Learned about the responsibilities of a DBA in a professional environment.
- Understood the relationship between database servers, applications, networks, and storage.
- Connected the theoretical concepts learned during the internship with real-world IT infrastructure.
- Gained practical awareness of professional database administration.

1.10 Conclusion

The Data Center visit was an important learning activity during the internship. It provided practical exposure to a professional IT infrastructure and helped us understand how different types of servers and supporting systems work together.

During the visit, an Oracle Database Server was also observed, and its basic working and importance were explained. The role of the DBA in maintaining database availability, security, performance, connectivity, and storage was also discussed.

Overall, the visit helped bridge the gap between theoretical DBA concepts and their practical implementation in a real-world Data Center environment.

1.11 Data Center Visit Screenshots

The following photographs/screenshots can be added as evidence of the Data Center visit.

Figure 1.1: Data Center Overview



Figure 1.2: Servers Installed in the Data Center



Figure 1.3: Oracle Database Server



Figure 1.4: Server Rack and Data Center Infrastructure



Part 2 — DBA Challenges, Errors and Solutions

2.1 Introduction to Internship Challenges

During the internship, various technical challenges and problems were encountered while working with Oracle Database, Oracle Linux, VMware, SQL Developer, and TOAD.

These challenges provided practical experience in database administration and troubleshooting. Some problems were related to database connectivity, Oracle Listener, TNS configuration, user privileges, network connectivity, Linux environment configuration, and database storage.

Each problem required identifying the cause, applying an appropriate solution, and verifying that the database environment was working correctly.

The troubleshooting experience helped in developing practical DBA skills and improved understanding of Oracle Database administration.

2.2 Common DBA Problems and Errors

During the internship, several common database administration problems and errors were encountered or studied. The following table summarizes the major problems, their common causes, and possible solutions.

No.	Possible Issue	Common Cause	Possible Solution
1	**TOAD cannot connect to database**	Host, port, or service name is incorrect	Check host/IP address, port 1521, service name, and TNS configuration
2	**ORA-12514: Listener does not currently know of service requested**	Requested database service is not registered with the listener	Check lsnrctl services and verify database service registration
3	**ORA-12154: TNS could not resolve the connect identifier specified**	TNS alias cannot be resolved	Check tnsnames.ora, TNS alias, and run tnsping
4	**ORA-12541: TNS: No Listener**	Oracle Listener is not running	Check lsnrctl status and start the listener
5	**ORA-01017: Invalid username/password**	Incorrect username or password	Verify the credentials or reset the password
6	**ORA-28000: The account is locked**	Database user account is locked	Unlock the database account
7	**ORA-28001: The password has expired**	User password has expired	Reset the user's password
8	**Insufficient Privileges**	User does not have the required privilege	Grant the required system or object privilege
9	**Tablespace Full / Insufficient Space**	Tablespace has insufficient free space	Check tablespace usage and extend the required storage
10	**User Cannot Create Database Object**	Required privilege is missing	Grant the required system or object privilege
11	**TOAD Shows Wrong or Old Database Information**	Incorrect database connection or service	Verify and select the correct database connection
12	**PDB is Not Open**	PDB is mounted or closed	Open the required PDB in the appropriate mode
13	**Listener Service Missing After Database Restart**	Database service was not dynamically registered	Check database and listener configuration
14	**TOAD Freezes or Does Not Respond**	Network connection or database response is slow	Check network connectivity and database activity

No.	Possible Issue	Common Cause	Possible Solution
15	**SQL Execution Error in TOAD**	SQL syntax, object, or privilege problem	Check the error code and correct the SQL or privileges
16	**SQL*Plus Command Not Found**	Oracle environment variables or PATH are not configured	Configure ORACLE_HOME, ORACLE_SID, and PATH
17	**Oracle Environment Variables Not Set**	Oracle environment was not configured	Configure the required Oracle environment variables
18	**VMware Shared Folder Not Visible**	Shared folder or VMware configuration problem	Verify Shared Folder settings and mount the folder
19	**Permission Denied in Oracle Linux**	User does not have sufficient permissions	Use appropriate privileges and correct ownership/permissions
20	**YUM Repository Error**	Incorrect repository configuration	Correct the repository configuration and rebuild the YUM cache
21	**Could Not Resolve Host**	Network or DNS configuration problem	Check network connectivity and DNS configuration
22	**Oracle Linux Network Interface Not Working**	Network interface or NetworkManager configuration problem	Check the network interface and NetworkManager
23	**SQL Developer Connection Failure**	Incorrect host, port, service name, or listener configuration	Verify connection details and listener status
24	**Database Link Connection Problem**	Incorrect TNS configuration or remote connectivity problem	Check TNS configuration and remote database connectivity
25	**PDB Service Not Registered with Listener**	PDB service registration problem	Verify PDB service and listener registration
26	**Database Service Unavailable After Restart**	Database or required service was not started	Start the required database service and verify listener registration
27	**Incorrect Host or IP Address**	Wrong database server address	Verify the database server IP address
28	**Port 1521 Connection Failed**	Listener or network/firewall problem	Check listener status and network connectivity
29	**TNSPING Failed**	TNS alias or network configuration problem	Check tnsnames.ora, listener, and network connectivity
30	**Database Instance Not Open**	Database is mounted or not completely started	Start/open the database and verify instance status

2.3 Oracle Database Connectivity Issues

Database connectivity was one of the important areas during the internship. Oracle Database was accessed using different client tools such as **SQL Developer and TOAD**.

Several factors can affect Oracle connectivity, including:

- Incorrect hostname or IP address
- Incorrect port number
- Incorrect service name
- Listener not running
- Database service not registered
- Incorrect TNS configuration
- Network connectivity problems
- Incorrect username or password

The Oracle Listener normally uses port **1521** for client connections. Therefore, the listener status and required database services must be verified when a connection problem occurs.

Commands commonly used for connectivity troubleshooting include:

lsnrctl status

lsnrctl services

and:

tnsping <service_name>

These commands help the DBA identify whether the listener, service, and TNS configuration are working correctly.

2.4 Oracle Listener and Service Issues

The Oracle Listener is an important component of Oracle Database connectivity. It receives connection requests from clients and directs them to the appropriate database service.

During Oracle administration, problems may occur when the listener is stopped or when the required database service is not registered.

Common listener-related errors include:

- **ORA-12514**
- **ORA-12541**
- Listener supports no services
- Database service missing from lsnrctl services
- Service not registered after database restart

The following commands can be used to troubleshoot listener-related problems:

lsnrctl status

lsnrctl services

lsnrctl start

The database and listener service registration should then be verified to ensure that clients can connect successfully.

2.5 TNS and Network Connectivity Issues

TNS configuration is important when connecting to Oracle Database using a network service name.

A common TNS-related error encountered during Oracle administration is:

ORA-12154: TNS could not resolve the connect identifier specified

This usually occurs when the specified TNS alias cannot be resolved.

The tnsnames.ora file should therefore be checked to ensure that the correct database service information is configured.

The following command can be used to test TNS connectivity:

tnsping <service_name>

Network-related problems may also occur because of incorrect IP addresses, network interfaces, DNS configuration, firewall restrictions, or unavailable ports.

Therefore, both network connectivity and Oracle configuration should be checked during troubleshooting.

2.6 SQL Developer and TOAD Connection Issues

SQL Developer and TOAD were used as database client tools during the internship.

Connection problems in these tools can occur due to incorrect:

- Username
- Password
- Host/IP address
- Port
- Service name
- TNS alias
- Listener configuration

For example, an **ORA-01017** error indicates an authentication problem, while **ORA-12514** generally indicates that the listener does not recognize the requested database service.

The connection information should therefore be verified before troubleshooting the database itself.

After correcting the configuration, the connection should be tested again from SQL Developer or TOAD.

2.7 Database User and Privilege Issues

Database security and user management are important responsibilities of a DBA.

A user may face problems when:

- The account is locked.
- The password has expired.
- Required privileges are missing.
- The user cannot create database objects.
- The user cannot access another user's table.

For example, insufficient privileges can prevent a user from creating tables or accessing specific database objects.

The DBA can resolve such problems by reviewing the user's roles and privileges and granting only the permissions required for the user's responsibilities.

This follows the **Principle of Least Privilege**, where users should receive only the permissions they actually need.

2.8 Database Storage and Tablespace Issues

Database storage is another important area of DBA administration.

Oracle Database uses tablespaces to logically organize database storage. If a tablespace becomes full or has insufficient free space, database operations may be affected.

A DBA should regularly monitor:

- Tablespace status
- Available space
- Used space
- Datafiles
- Storage growth

When insufficient storage is identified, the DBA can take appropriate action, such as extending an existing datafile or adding additional storage, depending on the database configuration.

Regular storage monitoring helps prevent unexpected database problems.

.9 Oracle Linux and Environment Configuration Issues

Oracle Database administration also required working with Oracle Linux.

Some problems were related to the Oracle environment and Linux configuration, including:

- sqlplus command not found
- ORACLE_HOME not configured
- ORACLE_SID not configured
- Incorrect PATH
- Permission denied
- VMware shared folder not visible
- YUM repository configuration problems
- Network interface problems

Oracle environment variables are important because Oracle utilities such as SQL*Plus and Listener tools depend on the correct Oracle environment.

The oraenv utility can be used to configure the Oracle environment:

. oraenv

The required Oracle SID can then be specified.

Correct Linux permissions, network configuration, repository configuration, and Oracle environment settings are important for smooth database administration.

2.10 Troubleshooting Summary

The troubleshooting process followed during the internship generally consisted of the following steps:

1. Identify the error or problem.
2. Read and understand the Oracle error message.
3. Identify the possible cause.
4. Check the relevant Oracle or Linux configuration.
5. Apply the appropriate solution.
6. Restart or reconfigure the required service if necessary.
7. Verify the configuration.
8. Test the database connection or operation again.

This systematic troubleshooting approach helped in resolving database connectivity, listener, TNS, user, privilege, network, and Linux configuration problems.

2.11 Learning Outcomes

After working with these challenges, the following practical skills were developed:

- Understanding Oracle Database error messages.
- Troubleshooting database connectivity problems.
- Managing Oracle Listener services.
- Working with TNS configuration.
- Testing network and database connectivity.
- Using SQL Developer and TOAD for database access.
- Managing database users and privileges.
- Understanding PDB and database service registration.
- Monitoring database storage and tablespaces.
- Configuring Oracle environment variables in Linux.
- Troubleshooting Linux permissions and network issues.
- Developing a systematic approach to DBA problem-solving.

These experiences improved practical understanding of Oracle Database administration beyond theoretical concepts.

2.12 Conclusion

During the internship, different challenges related to Oracle Database, Oracle Linux, networking, database connectivity, Listener services, TNS configuration, SQL Developer, TOAD, users, privileges, and storage were encountered and studied.

Troubleshooting these problems provided practical experience in identifying errors, finding their causes, applying appropriate solutions, and verifying the results.

Overall, the challenges faced during the internship significantly improved understanding of real-world DBA responsibilities and demonstrated the importance of systematic troubleshooting in maintaining a reliable, secure, and available Oracle Database environment.